PROJECT 2:
Hear: Time and Data

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DESN22848 Interaction Design: Media, Motion and the Body
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Table of contents

(Table of contents)

01 | Design Statement

02 | Research

03 | Formstorming

04 | Reflection

05 | Citations

06 | Appendix (WEEK6-9)

Design Statement: Concept

This project aims to explore transforming intangible sound energy into dynamic, interactive visual forms. It seeks to establish a web-based digital environment built on p5.js that responds to audio input in real time. The core concept is to create an immersive experience where sound becomes a tangible, visible medium that users can observe and influence.

Sound is presented through four distinct visual “environments” or modes. Using a microphone, users can directly control these geometric patterns with their own voice, music, or any ambient sound, causing them to instantly breathe, deform, and react.

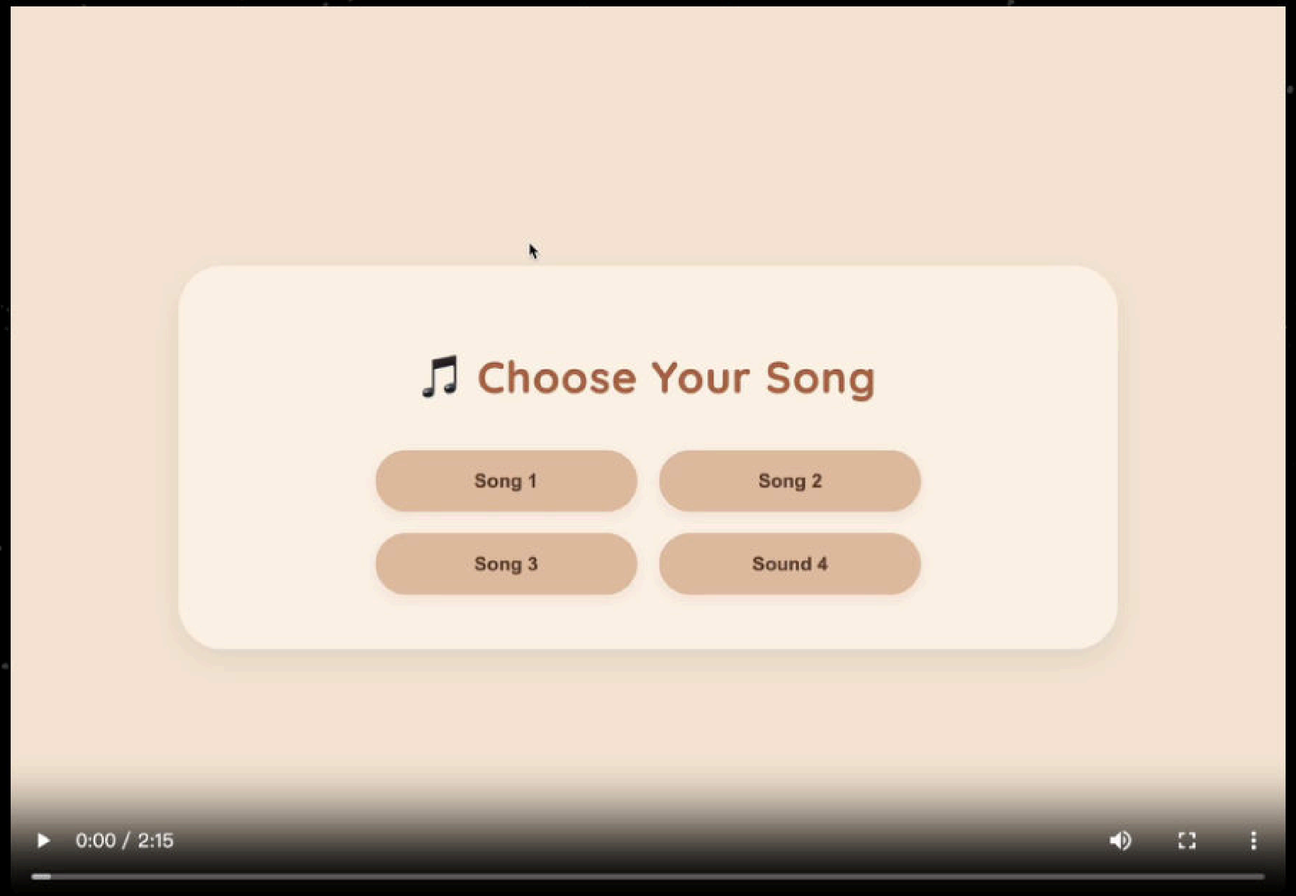
Design Principles:

- Real-Time Interactivity: Primary feedback loops are immediate. Changes in audio input (volume, frequency) instantly trigger recognizable shifts in visual output, establishing a direct and intuitive connection between action and reaction.

- Multi-Modal Representation: Acknowledging that sound can be interpreted in multiple ways, four distinct visual modes are provided. This allows users to find a visual language that resonates with their personal experience of the audio.

- Abstract expression: By using abstract geometric shapes instead of concrete images, users can perceive the essence of sound more purely.

Design Statement: Final Project 2 Design



(Due to software issues, the video cannot be imported into Figma.)

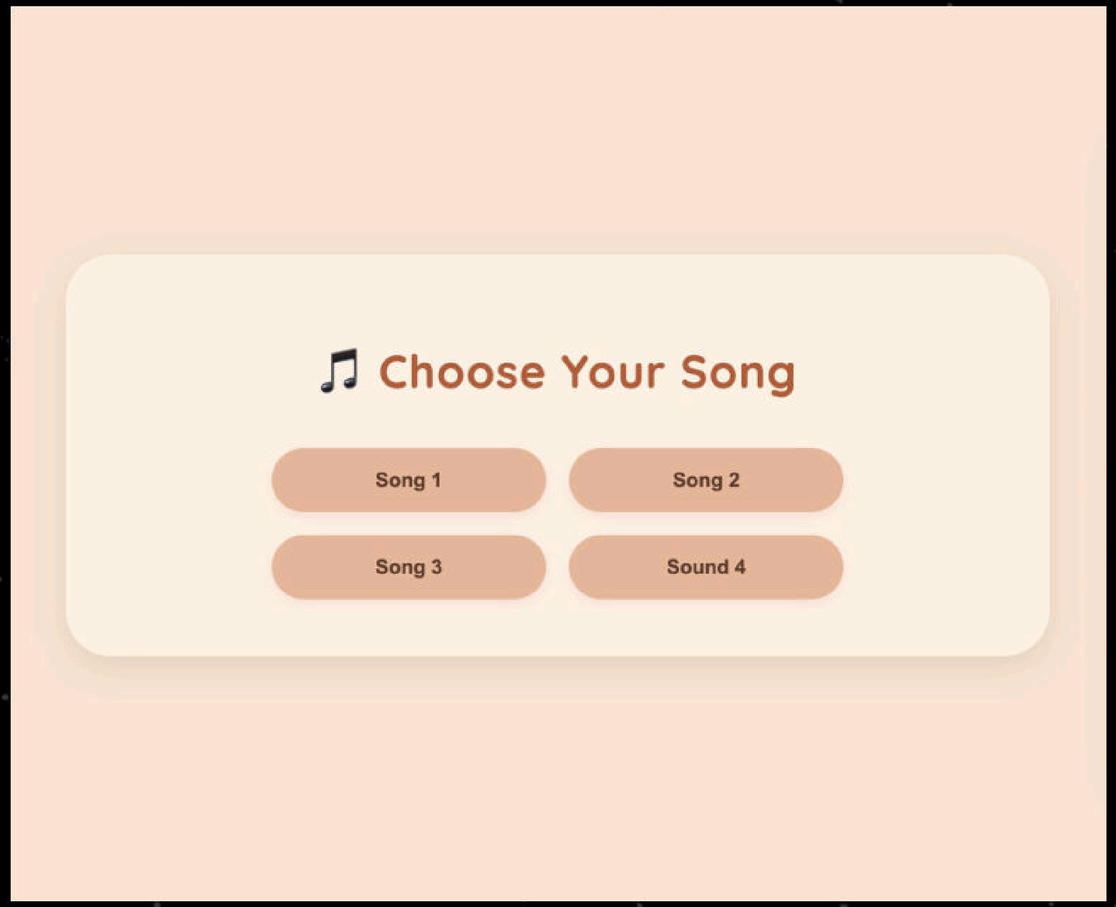
URL:
http://zhou37.phoenix.sheridanc.on.ca/DESN22848/Sound_Header_Template/

YOUTUBE:
https://youtu.be/AmP7iFSQApk

Technology:

- P5.js
- Music
- Microphone
- HTML / CSS / JS

Design Statement: Introduction

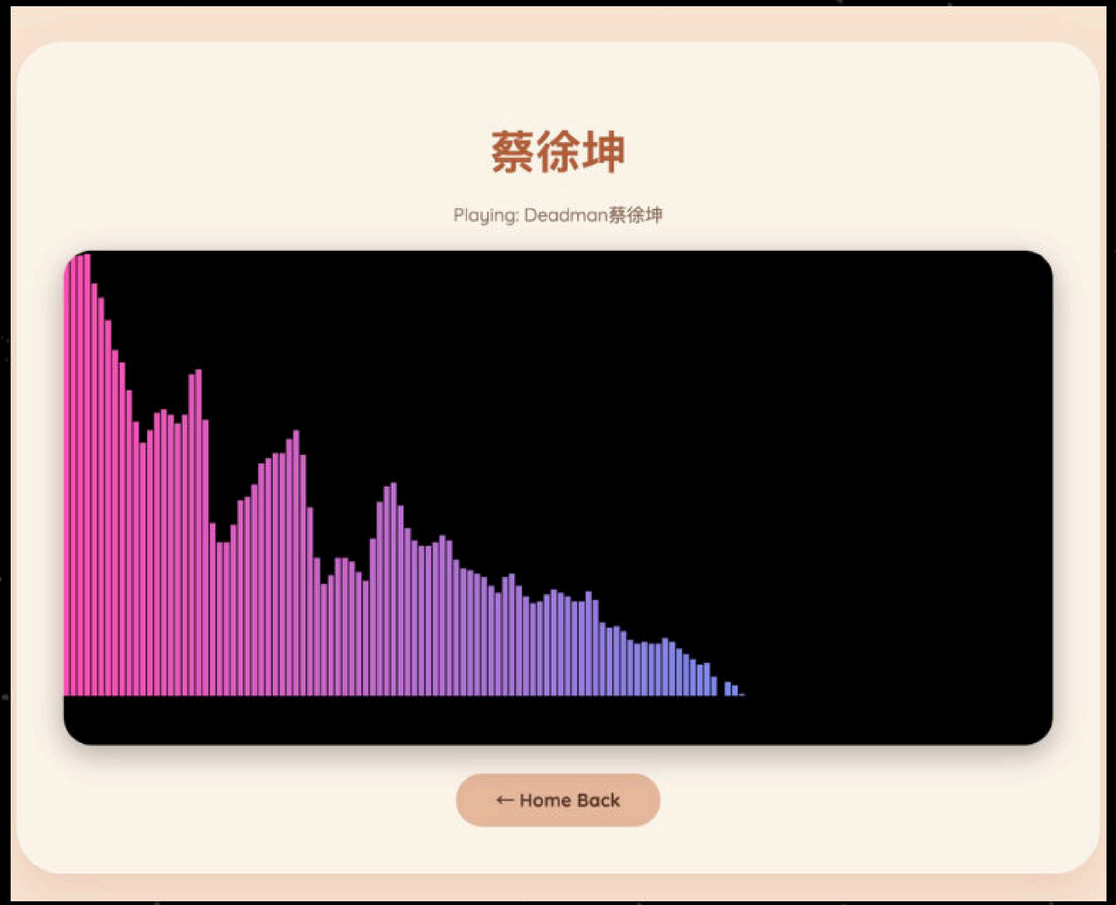


Main page

The four buttons in the center of the page correspond respectively to "song1, song2, song3, sound4". Each button is accompanied by a title. Clicking on it will redirect you to the designated subpage.

The first three belong to the visual effects produced by built-in music. The fourth one belongs to the visual effects produced by external sounds.

Design Statement: Introduction



Song 1:

Create visual effects through built-in music. The visual effects will change, undulate and stretch along with the rhythm and volume of the music.

```
function setup() {
    let container = document.getElementById('canvasContainer');
    let canvas = createCanvas(container.offsetWidth,
    container.offsetHeight);
    canvas.parent('canvasContainer');

    fft = new p5.FFT(0.8, 128);
    fft.setInput(song);
}

function draw() {
    background(0);

    if (isPlaying) {
        let spectrum = fft.analyze();
        noStroke();

        let barWidth = width / spectrum.length;
        let c1 = color(255, 80, 180);
        let c2 = color(80, 160, 255);

        for (let i = 0; i < spectrum.length; i++) {
            let value = spectrum[i];
            let h = map(value, 0, 255, 0, height);
            let t = i / (spectrum.length - 1);
            let c = lerpColor(c1, c2, t);

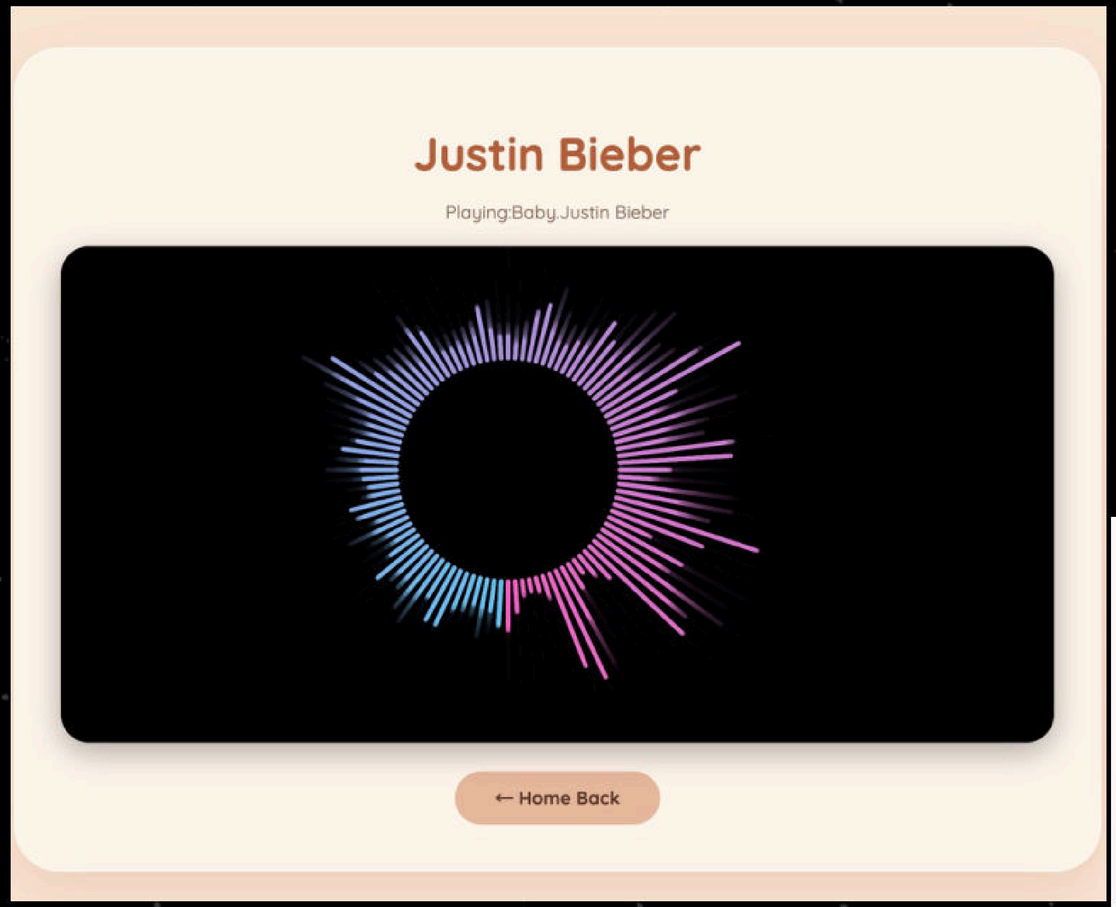
            fill(c);
            rect(i * barWidth, height - h, barWidth - 1, h);
        }
    }
}
```

```
    function mousePressed() {
        if (mouseX > 0 && mouseX < width && mouseY > 0 && mouseY < height) {
            if (loadError) {
                alert("The music file cannot be read. Please check if the
                file name is Baby (Justin Bieber).mp3 and make sure to run it
                using VS Code Live Server.");
                return;
            }

            if (song.isLoaded()) {
                if (!isPlaying) {
                    userStartAudio();
                    song.loop();
                    isPlaying = true;
                    document.getElementById('clickHint').style.display =
                    'none';
                } else {
                    song.pause();
                    isPlaying = false;
                    document.getElementById('clickHint').style.display =
                    'block';
                    document.getElementById('clickHint').innerText =
                    "PAUSED";
                }
            }
        }
    }

    function windowResized() {
        let container = document.getElementById('canvasContainer');
        resizeCanvas(container.offsetWidth, container.offsetHeight);
    }
</script>
</body>
```

Design Statement: Introduction



Song 2:

Create visual effects through built-in music. The visual effects will change, undulate and stretch along with the rhythm and volume of the music.

The most fundamental visual effects are very suitable to appear when listening to R&B.

```
<script>
    let song, fft, amp;
    let isPlaying = false;
    let radius, number;

    function preload() {
        song = loadSound('Baby.Justin Bieber.MP3');
    }

    function setup() {
        let container = document.getElementById('canvasContainer');
        let canvas = createCanvas(container.offsetWidth,
        container.offsetHeight);
        canvas.parent('canvasContainer');

        amp = new p5.Amplitude();
        fft = new p5.FFT(0.8, 1024);
        fft.setInput(song);
        amp.setInput(song);

        radius = height * 0.25;
        number = 100;
        strokeCap(ROUND);
    }

    function draw() {
        background(0, 30);

        if (isPlaying) {
            let spectrum = fft.analyze();
            let level = amp.getLevel();

            translate(width/2, height/2);
```

```
function mousePressed() {
    if (mouseX > 0 && mouseX < width && mouseY > 0 && mouseY < height) {
        if (song.isLoaded()) {
            if (!isPlaying) {
                userStartAudio();
                song.loop();
                isPlaying = true;
                document.getElementById('clickHint').style.display =
                'none';
            } else {
                song.pause();
                isPlaying = false;
                document.getElementById('clickHint').style.display =
                'block';
                document.getElementById('clickHint').innerText =
                "PAUSED";
            }
        }
    }
}

function windowResized() {
    let container = document.getElementById('canvasContainer');
    resizeCanvas(container.offsetWidth, container.offsetHeight);
    radius = height * 0.25;
```

Design Statement: Introduction



Song 3:

Create visual effects through built-in music. The visual effects will change, undulate and stretch along with the rhythm and volume of the music.

Through the rotation and appearance of objects, as well as music, I hope to present that feeling of interstellar travel.

```
function setup() {
    let container = document.getElementById('canvasContainer');

    let canvas = createCanvas(container.offsetWidth,
    container.offsetHeight, WEBGL);
    canvas.parent('canvasContainer');

    amp = new p5.Amplitude();
    amp.setInput(song);
}

function draw() {
    background(0);

    if (isPlaying) {

        rotateX(frameCount * 0.01);
        rotateY(frameCount * 0.01);

        let level = amp.getLevel();

        if (level > 0.02) {
            let spawnCount = floor(map(level, 0.02, 0.3, 2, 8));
            for (let i = 0; i < spawnCount; i++) {
                let angle = random(TWO_PI);
                let speed = random(2, 5) * level * 50;
                boxes.push({
                    x: 0, y: 0, z: 0, //

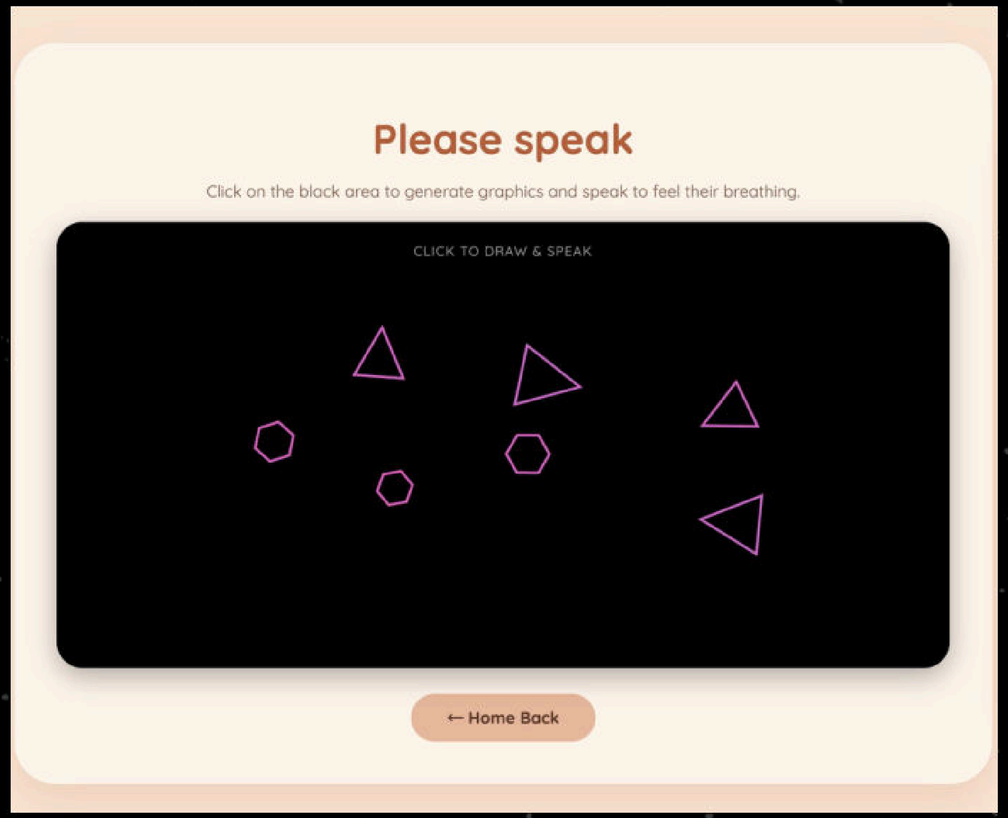
                    vx: cos(angle) * speed,
                    vy: sin(angle) * speed,
                    vz: random(-1, 1) * speed,
                    size: random(10, 40) * (1 + level * 2),
                    alpha: 255,
```

```
    function mousePressed() {

        if (mouseX > 0 && mouseX < width && mouseY > 0 && mouseY < height) {
            if (song.isLoaded()) {
                if (!isPlaying) {
                    userStartAudio();
                    song.loop();
                    isPlaying = true;
                    document.getElementById('clickHint').style.display =
                    'none';
                } else {
                    song.pause();
                    isPlaying = false;
                    document.getElementById('clickHint').style.display =
                    'block';
                    document.getElementById('clickHint').innerText =
                    "PAUSED";
                }
            }
        }
    }

    function windowResized() {
        let container = document.getElementById('canvasContainer');
        resizeCanvas(container.offsetWidth, container.offsetHeight);
    }
</script>
```

Design Statement: Introduction



Sound 4:

Create visual effects through external sounds. The visual effects will change, undulate and stretch along with the rhythm and volume of the sound. However, the objects in the visual effects are generated by the user's clicks.

```
<script>
    let mic;
    let shapes = [];
    let micStarted = false;

    function setup() {
        let container = document.getE
        let canvas = createCanvas(con
        container.offsetHeight);
        canvas.parent('canvasContaine

        mic = new p5.AudioIn();
    }

    function draw() {
        background(0);

        let level = 0;
        if (micStarted) {
            level = mic.getLevel();
        }

        for (let s of shapes) {
            s.update(level);
            s.display();
        }

        if (shapes.length === 0) {
            fill(255, 50);
            textAlign(CENTER);
            text("Click here to start creating", width/2, height/2);
        }
    }

function mousePressed() {

    if (mouseX > 0 && mouseX < width && mouseY > 0 &&
        if (!micStarted) {
            userStartAudio();
            mic.start();
            micStarted = true;
        }

        shapes.push(new GeoShape(mouseX, mouseY));
    }
}

class GeoShape {
    constructor(x, y) {
        this.x = x;
        this.y = y;
        this.base = random(30, 60);

        this.size = this.base;
        this.type = floor(random(3));

        this.rot = random(TWO_PI);
        this.rotSpeed = random(-0.02, 0.02);
    }

    update(level) {

        let growth = map(level, 0, 0.25, 0, 200, true)
        this.size = this.base + growth;
        this.rot += this.rotSpeed;
    }

        push();
        translate(this.x, this.y);
        rotate(this.rot);

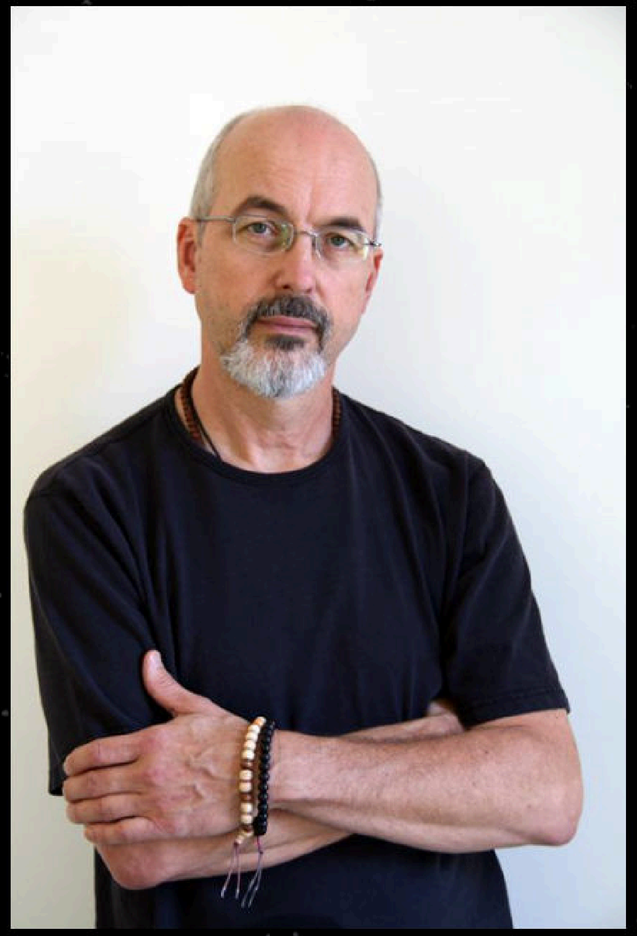
        if (this.type === 0) {
            this.triangleShape();
        } else if (this.type === 1) {
            rectMode(CENTER);
            rect(0, 0, this.size, this.size);
        } else {
            this.polygon(6, this.size);
        }
        pop();
    }

    triangleShape() {
        beginShape();
        vertex(0, -this.size / 2);
        vertex(-this.size / 2, this.size / 2);
        vertex(this.size / 2, this.size / 2);
        endShape(CLOSE);
    }

    polygon(n, r) {
        beginShape();
        for (let i = 0; i < n; i++) {
            let a = TWO_PI * i / n;
            vertex(cos(a) * r / 2, sin(a) * r / 2);
        }
        endShape(CLOSE);
    }
}

function windowResized() {
    let container = document.getElementById('canvasContainer');
    resizeCanvas(container.offsetWidth, container.offsetHeight);
```

Research: Bill Viola《The Messenger》(1996)



URL: youtu.be/xArFDtN3B2M — Bill Viola: The Messenger (1996) | Durham Cathedral, England

Basic Information:

- **Artist: Bill Viola (USA, 1951–)**
- **Title: The Messenger**
- **Year Created: 1996**
- **Commissioned by Durham Cathedral, UK, as part of the "Year of Visual Arts in Britain" celebrations, it premiered at Durham Cathedral. Subsequently toured globally, including the Art Gallery of New South Wales, Sydney (1998) and Fruitmarket Gallery, Edinburgh (1997).**

Description:

The Messenger is a 28 minute looping single-channel video installation. The image depicts a naked man fully submerged in a dark pool. He slowly rises to the surface, breaks through the water, takes a deep breath, and opens his eyes. He then gradually sinks back down, disappearing once more into the deep blue void. This sequence repeats four times within the 28 minute loop. The work is accompanied by ambient sound gentle underwater sounds gradually intensify to a crescendo as the figure surfaces, then fade back into tranquility.

When exhibited at Durham Cathedral, the work was projected onto the rear doors; within gallery spaces, viewers must traverse a pitch-black room to reach the screen. This extreme darkness initially induces a profound sense of disorientation, followed by a comforting relief upon the screen's appearance.

Research: 《The Messenger》: Information Commentary (Can be skipped)



Loop and Time:
Since her first university exhibition, Viola has consistently employed looped playback. This not only allows latecomers to view the work in its entirety but also establishes an analogy with “ritual” a regularly recurring, predictable action enabling individuals to enter an altered state of mind. Thus, The Messenger becomes a metaphor for life's cycles.

The deliberately slow pace “stretches” time. This treatment compels viewers to focus on subtle shifts through sustained observation how bubbles form halos around the figure's head, how ripples flow across the body. This directly resonates with Week 6's core concept: “Hear: Time and Data” (sound as a data stream unfolding over time).

Research: 《The Messenger》: Information Commentary (Can be skipped)



The Body as a Perceptual Medium:
Viola repeatedly emphasizes that art and knowledge should be “sensually experienced,” not confined to purely intellectual realms. Critic Valentina Valentini notes that Viola's imagery possesses a powerful force to awaken the bodies of Western audiences who have long been conditioned to a state where “the mind controls the body.”

The nude figures immersed in water within the work are not merely visual objects but active mediums of perception. The imagery of water evokes dual symbolism of birth (amniotic fluid) and death (returning to nature). In a conversation with Violant Porcel, Viola elaborated on his fascination with natural elements: “Nature also draws me spiritually: I seek to explore the universe's intrinsic laws in their transcendent dimensions to better understand humanity's habitat.”

Research: 《The Messenger》: Information Commentary (Can be skipped)



Emptiness/Silence as Material:
The alternation of darkness and light within the work creates a “reflective space.” Viewers must traverse a completely darkened room to reach the screen the extreme darkness initially induces a profound sense of disorientation, which later transforms into a comforting intimacy when the screen appears. This resonates with Nam June Paik's concept in Zen for Film (1964) that “emptiness/silence is also material”—darkness and silence are equally vital components of visual language.

Research: 《The Messenger》: Information Commentary (Can be skipped)



Spiritual Pursuit and Trans-Religious Vision:
Viola's readings span Zen Buddhism, Sufism, and Christian mysticism, presenting a vision of human experience that transcends specific religions. One of his favorite authors is John of the Cross, the imprisoned monk who, instead of responding with hatred, wrote The Spiritual Canticle "whose verses possess boundless expressive power. When I read it, I understood the effect art can produce." Viola believes: "Religion and art are similar; both focus on the essence of humanity."

Research: 《The Messenger》: Inspiration for My Project



- The concept of cyclical structures inspired me to introduce “cyclic memory” into sound visualization, allowing the sounds from the past few seconds to influence the current visuals and create richer temporal layers.
- The idea of bodily perception reinforced my understanding that “the user's voice is an extension of their physical movements”, sound is a form of the body's presence.
- The alternation of darkness and light revealed to me that silence and gaps are equally vital components of visual language; the frame need not be filled at all times.

Research: Granular-Synthesis《Modell 5》(1994-1996)



URL: youtu.be/tvZ1oZea7C8 Granular-Synthesis: Modell 5 (1994)

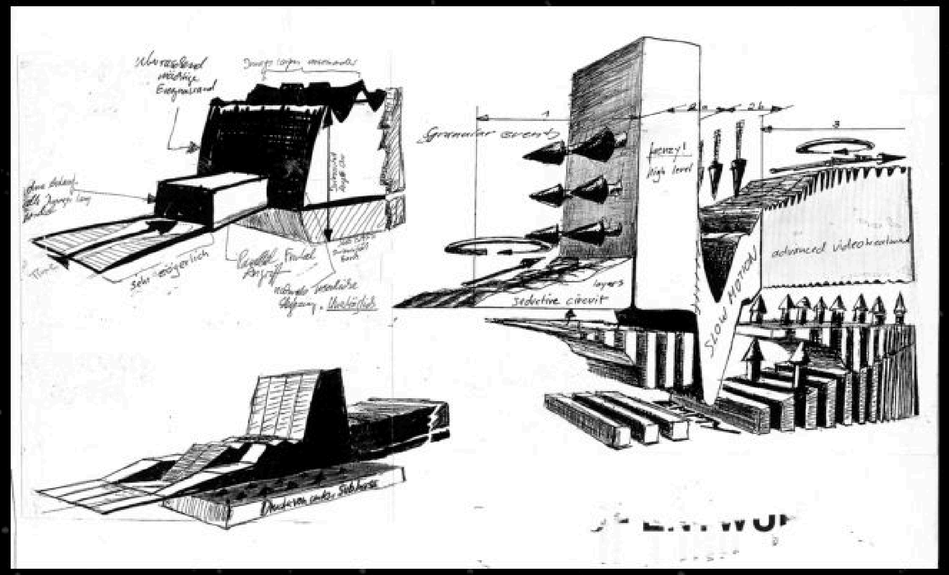
Basic Information:

- Artist Collective: Granular-Synthesis (comprising Kurt Hentschläger and Ulf Langheinrich, active in Vienna from 1991 to 2003)
- Work Title: Modell 5 (Subtitle: MotionControl)
- Creation Year: 1994–1996
- Work Type: Live audio-visual performance/installation, four-channel video output, eight-channel audio output
- Subject: Japanese performance artist Akemi Takeya

Work Description:
Modell 5 features Japanese performance artist Akemi Takeya as its subject, digitally processing her pre-recorded video and audio materials. The work presents Takeya's face repeated four times across four screens, with an image size of 10×3.5 meters. The artist collective's name “Granular-Synthesis” reveals its core methodology: decomposing recorded imagery into minimal information units for sampling and resynthesis.

During the performance, digitally processed visual and auditory elements emerge from four video channels and eight audio channels, interacting to create a multisensory perceptual experience. Deep bass frequencies and powerful mechanical rhythms dominate the entire space.

Research: 《Modell 5》: Information Commentary (Can be skipped)



Granularization:
Deconstruction and Reconstruction:
Originally a sound synthesis technique, Granular Synthesis extends this concept to the visual realm. In Modell 5, Takeya's visual and audio materials undergo an “analytical process” to separate them into the smallest processable units, which are then reassembled at different frequencies during the “reconstruction process.”

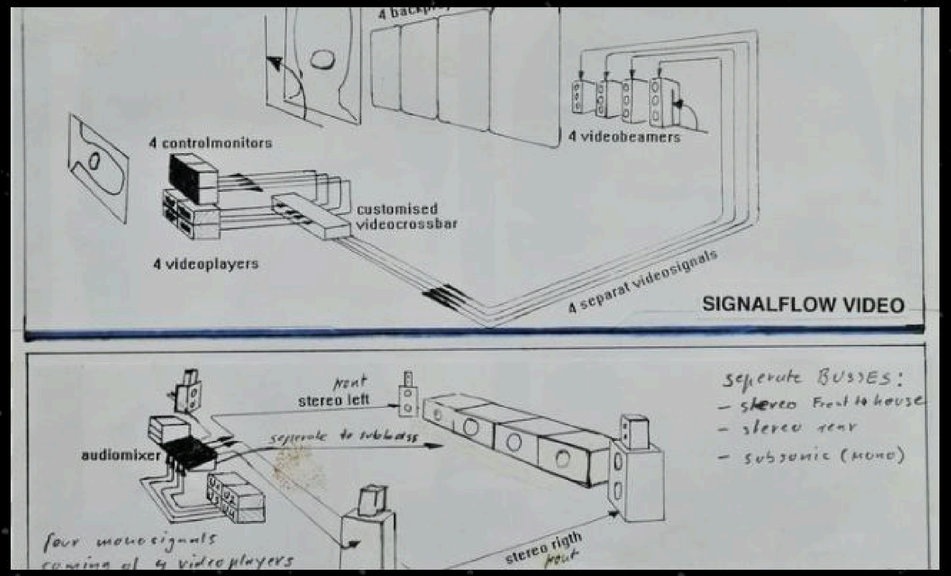
Critic Yvonne Spielmann observes:
“Through granular synthesis, new connections emerge—visual sequences interact with and become controllable by the auditory, yet the electronic processes of image and sound no longer synchronize conventionally: image and sound are separated, blurred, and perceived irregularly, like flickering.” This media reflexivity makes viewers aware that the variable organization of visual and auditory information constitutes a layer of the medium itself.

Research: 《Modell 5》: Information Commentary (Can be skipped)



Deconstruction of Audiovisual Synchrony:
In traditional video, the synchrony and stability of image and sound are essential. In Modell 5, these are dissolved. New connections are created visual sequences interact with and are simultaneously controlled by the auditory, yet the electronic processes of image and sound no longer synchronize in the conventional manner. The result: image and sound are separated, blurred, and perceived irregularly, like flickering.

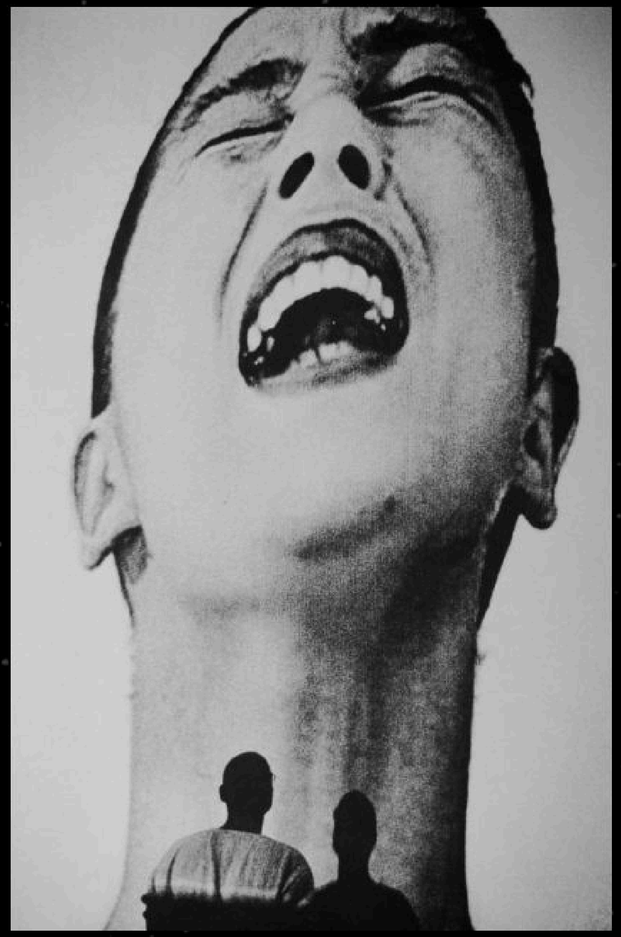
Research: 《Modell 5》: Information Commentary (Can be skipped)



Repetitive Intensity and Aesthetics of Violence: Viewers perceive this intervention in audio-visual material as “violent and painful,” as the artist dissects Takeya's voice and portrait: “Her natural rhythms are eliminated, replaced by mechanical rhythms within sequence control. Essentially, mathematical operations of digital analysis are applied to the video recording of Takeya's performance.”

Multiple critics draw parallels between this effect and Francis Bacon's paintings faces distorted and manipulated, only in the electronic realm this manipulation is achieved through the computer's mechanized processing. As Ars Electronica's review states: “The slow, blurred ‘growl’ effect in MODELL 5 does indeed evoke the appearance of Francis Bacon's paintings.”

Research: 《Modell 5》: Information Commentary (Can be skipped)



Physical Experience and Physiological Impact:
Granular Synthesis's choice of technology is not formalistic showmanship but grounded in a profound understanding of its potential: "The purpose of using this volume is not merely loudness, but because specific tones, pitches, and variations trigger physiological effects in the body, subsequently evoking deep psychological resonance. "Audiences are subjected to an intense barrage of stimuli, where attempts at rationalization are defeated by the work's sheer physicality this state of overload causes listeners to ‘lose self awareness to sink into the self,’ generating an experience that is 'disturbing, frightening, seductive, and erotic

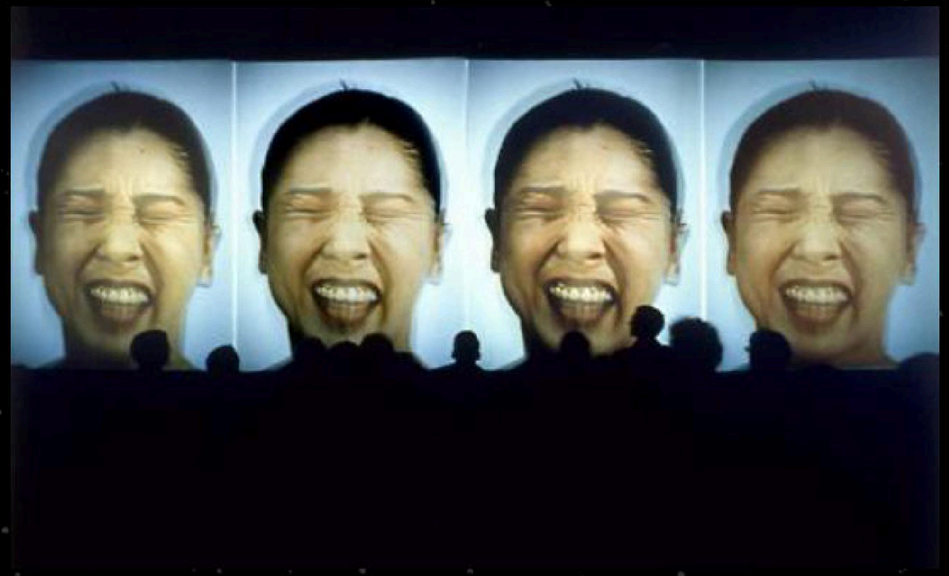
Research: 《Modell 5》: Information Commentary (Can be skipped)



Technical Development and Media Reflexivity:
To meet creative demands, Granular-Synthesis developed its proprietary software VARP 9: “No product on the market nor to this day met our requirements. In 1996, we sat down to design an audio-visual machine, integrating our experience from nonlinear video editing systems and audio sampling.” This technical development process itself embodies media reflexivity the work not only presents content but also exposes its production mechanisms.

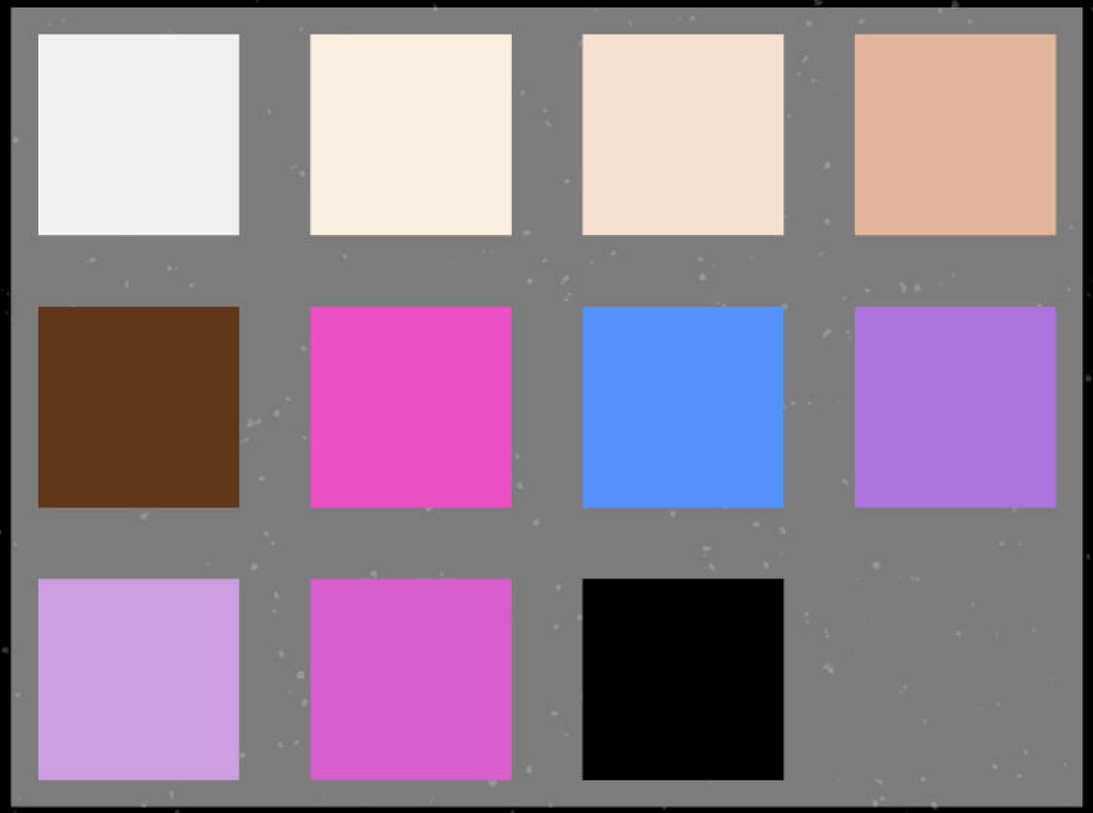
Historical Connection to Nam June Paik:
The work's overall abstract outcome resembles the “flicker effect” in video sequencers, yet differs in that its image sources are not linearly edited. Critics note: “Within the simultaneous auditory and visual variability, an impression of ‘visual musicalization’ emerges, structurally akin to Nam June Paik's early interventions with the waveform morphology of electronic media.”

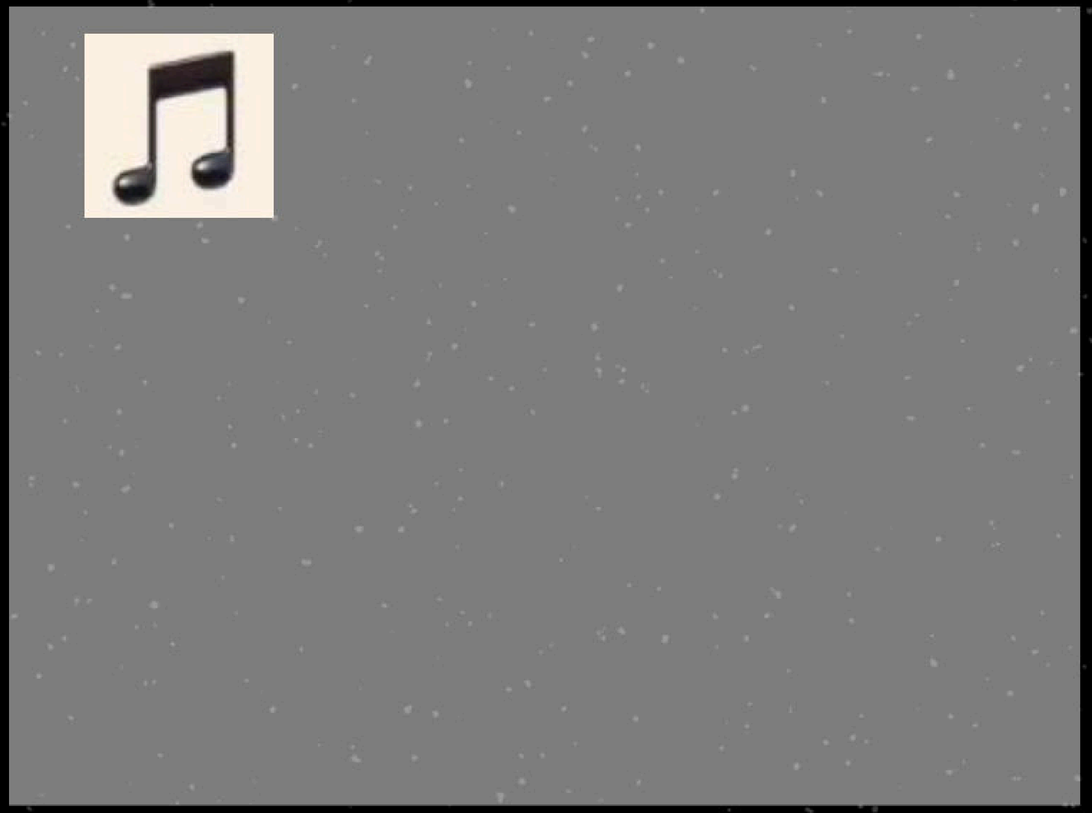
Research: 《Modell 5》: Inspiration for My Project



- Particle synthesis technology inspired me to consider: Could sound visualization also adopt a similar “sample-recombine” logic? Decomposing audio streams into minute units and resynthesizing visuals at different frequencies.
- The deconstruction of audio-visual synchronization challenges the singular “sound-drives-visuals” relationship in my p5.js project, I can explore more complex interactions between images and sound.
- The treatment of repetitive intensity suggests that rhythm and repetition wield powerful emotional influence in interaction design.
- The bodily experience dimension reminds me that interaction design is not merely visual, but also a physical, physiological experience.

Visual Research: Color and Icons





Visual Research: Font

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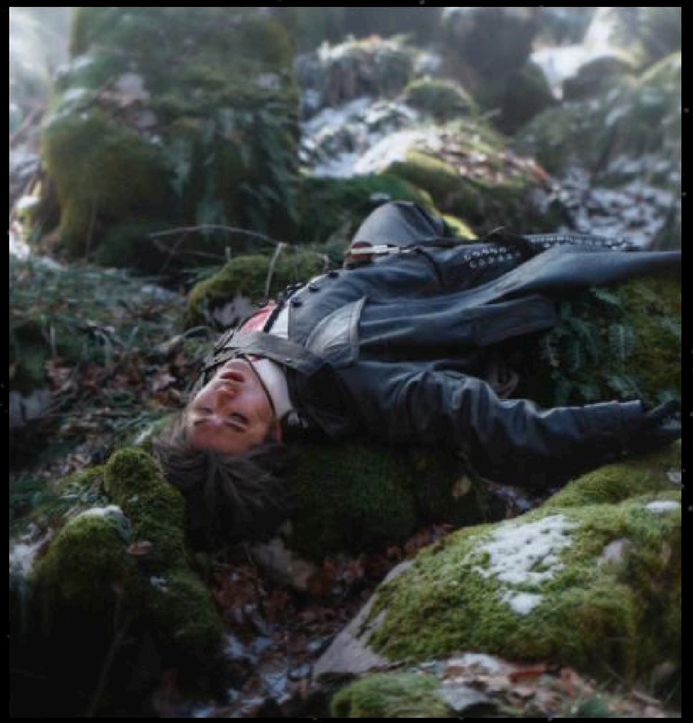
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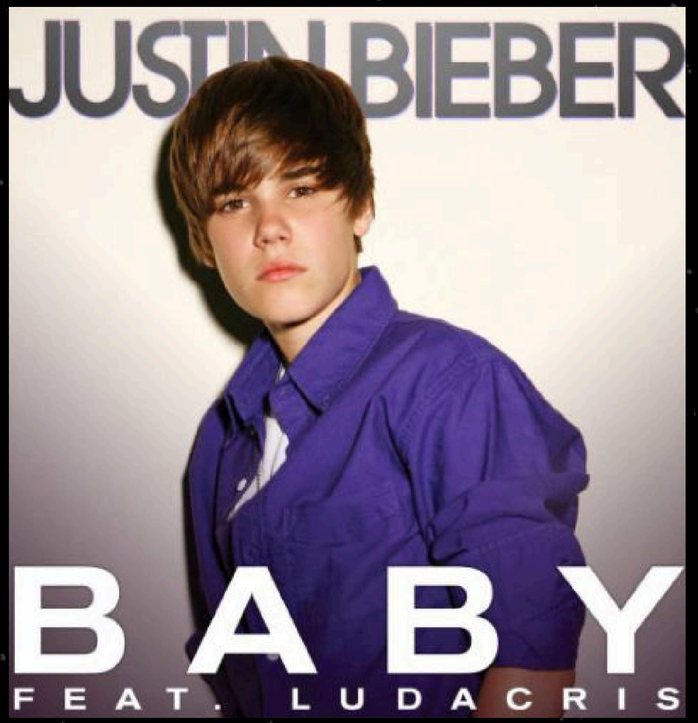
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Visual Research: Music



"Deadman" is a song performed by Xukun Cai. It was jointly created by Xukun Cai, Jacob Ray, Mark Mckenna and six other musicians. It was released globally as a single on May 28, 2025.

"Deadman" is a retro-style song, based on the soul music of the 1970s, integrating blues push-in techniques, Gospel harmonies and new psychedelic rock elements. The arrangement deliberately restores the rough texture of vinyl records from the 1950s and 1960s.



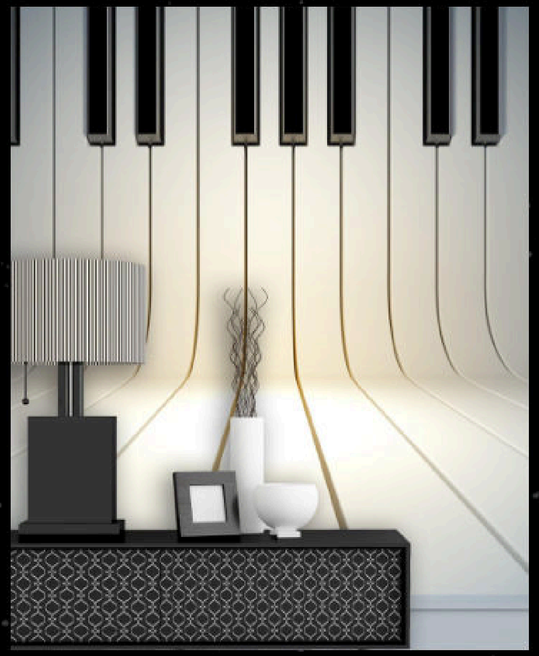
“Baby” is a teen pop dance song featuring Justin Bieber and Ludacris. It was written and composed by Justin Bieber, Ludacris, The Dream, Tricky Stewart, and Christina Milan.

The song is featured on Justin Bieber’s second studio album, *My World 2.0*, and was released as the album’s lead single on January 18, 2010, via Island Records.



"Star Crossing Night" was performed by Minghao Xu in collaboration with rapper GALI and was independently released as a digital single on August 30, 2025.
This song was co-written and composed by Minghao Xu. Its creative inspiration comes from the image of the night sky and fleeting meteors. The music integrates elements of UK Garage, Jersey Club and R&B.

Visual Research: Reference of research elements



Prototype 1:

For the elements, I referred to the keys of a piano. And among the songs I have selected and paired, there are a large number of instruments such as electric guitars, pianos, and basses. So I hope that the visual effect can well observe the "high-cut".



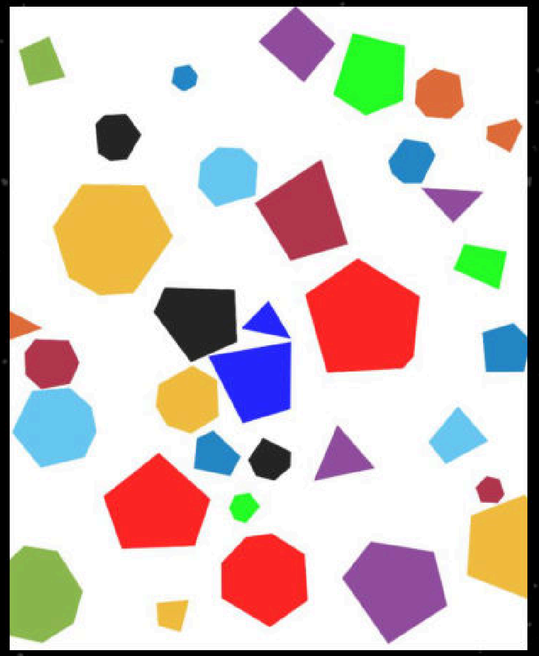
Prototype 2:

For the elements, I referred to an element from a cosmic scene in the movie "Interstellar". And the songs I selected to pair with, the creative inspiration of which came from the images of the night sky and fleeting meteors. So I hope to present that dizzy feeling of interstellar travel.



Prototype 3:

For the elements, I drew inspiration from the striking sensation of a drum kit. And among the songs I have selected and paired, there is a great deal of percussion. So I hope the visual effects can well present the changes in rhythm.ous

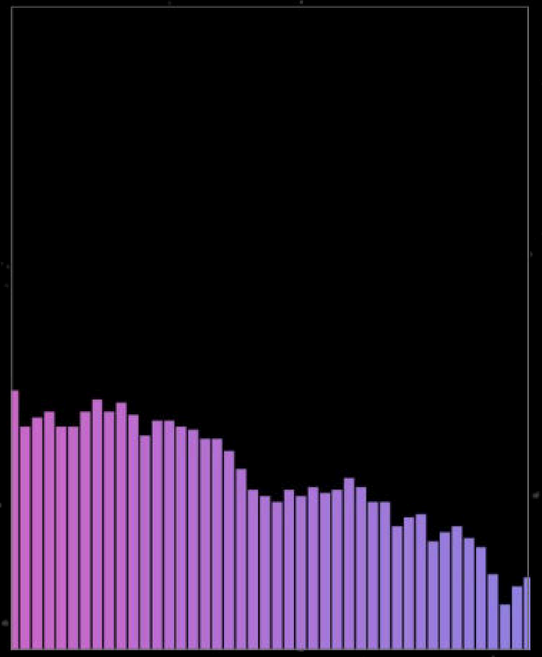


Prototype 4:

For elements, I drew inspiration from geometric shapes. Since the fourth visual effect changes through the sound of the microphone, I hope to create an interesting interactive mode in the visual effect.

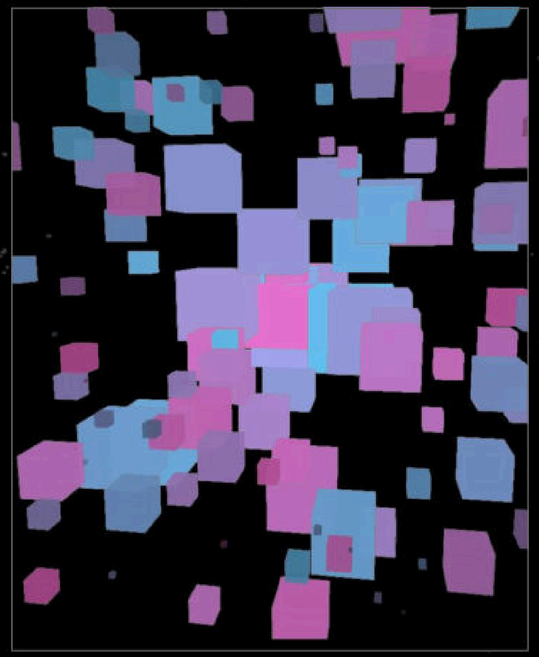
Visual Research: Technical Research (learn p5.js)

Inspired by the research of the two artists, I made some modifications to the four reactive visual environments established by activity2.



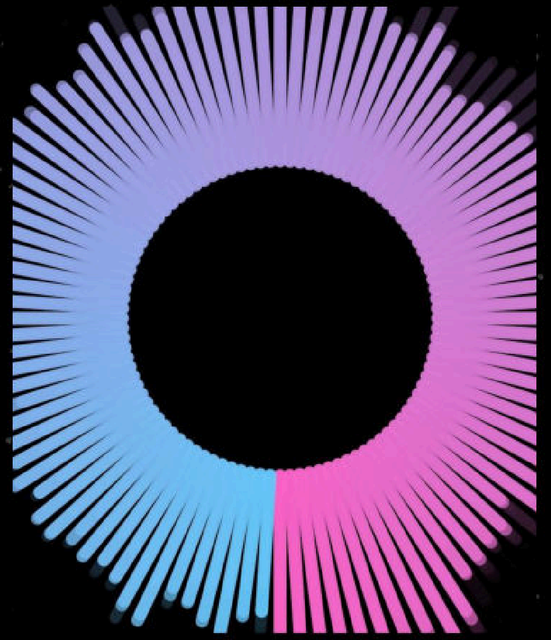
Prototype 1: Spectrogram

- Use fft.analyze to obtain the frequency array
- Result: I hope the visual effect can well observe the "High-cut". When simulating the effect of vinyl or old tapes, the high frequencies above 15kHz to 20kHz are usually cut off. In the spectral frequency graph, a distinct "blank" or dark area can be seen at the top, which is precisely the visual feature of the retro warm tone.



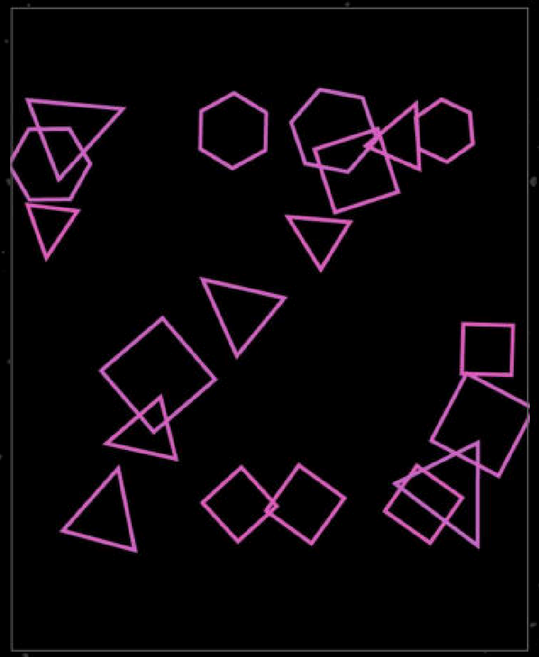
Prototype 2:3D geometric figure

Learn P5.js and try to build a cube of geometric shapes.
Result: I hope to create a sense of spatial immersion



Prototype 3: Basic Sound Input Test

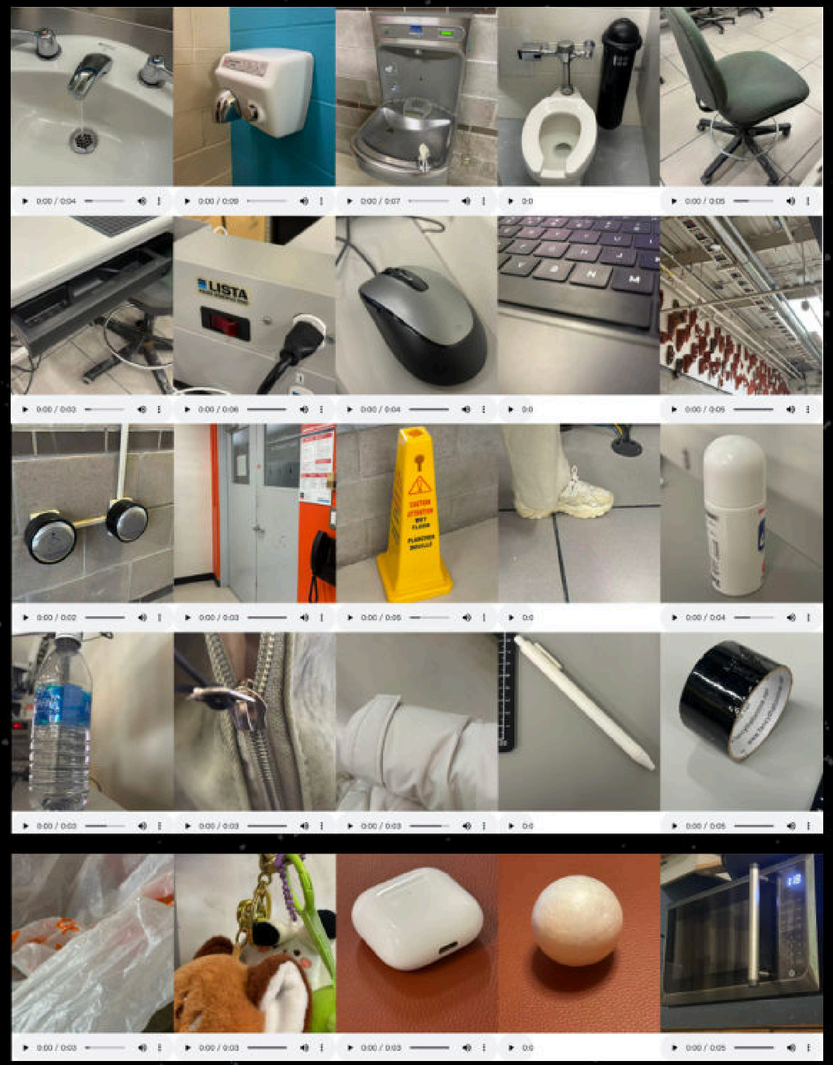
- Simple circle, volume control radius
- Result: Visuals too monotonous



Prototype 4: "Particle" Exploration

Inspired by Granular Synthesis, attempt particle effects
Result: The concept of "granulation" was successfully embodied, and the particle effect was presented using geometric shapes.

Formstorming

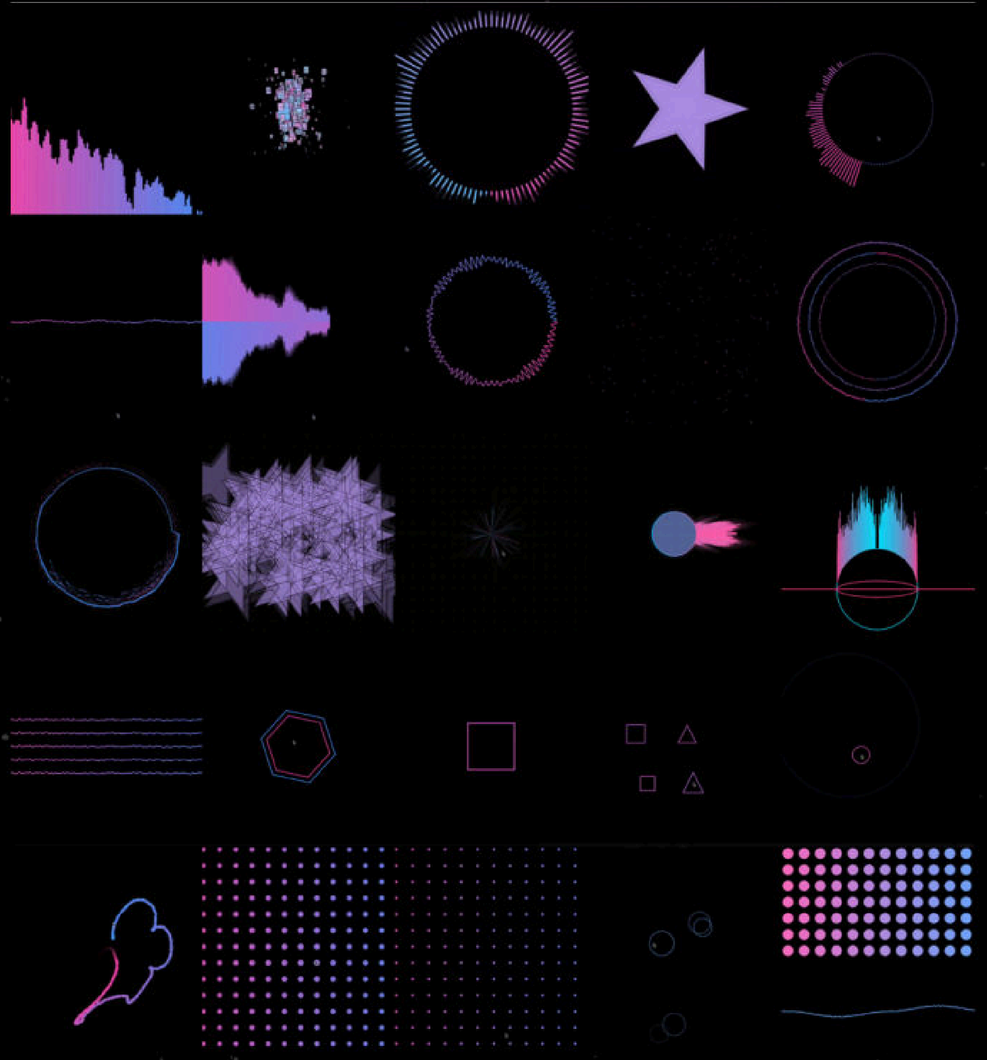


LINK: http://zhou37.phoenix.sheridanc.on.ca/DESN22848/MM&B%20Module%202%20Web%20Template/

Activity 1: Recording Everyday Sounds (Week 6)

Objective: Collect sounds encountered in daily life to build an audio library.
Process: Recorded 25 clips of everyday sounds using a phone.
Findings: Each sound possesses unique waveform and frequency characteristics.

Formstorming



LINK: http://zhou37.phoenix.sheridanc.on.ca/DESN22848/MM&B%20Module%202%20Web%20Template/

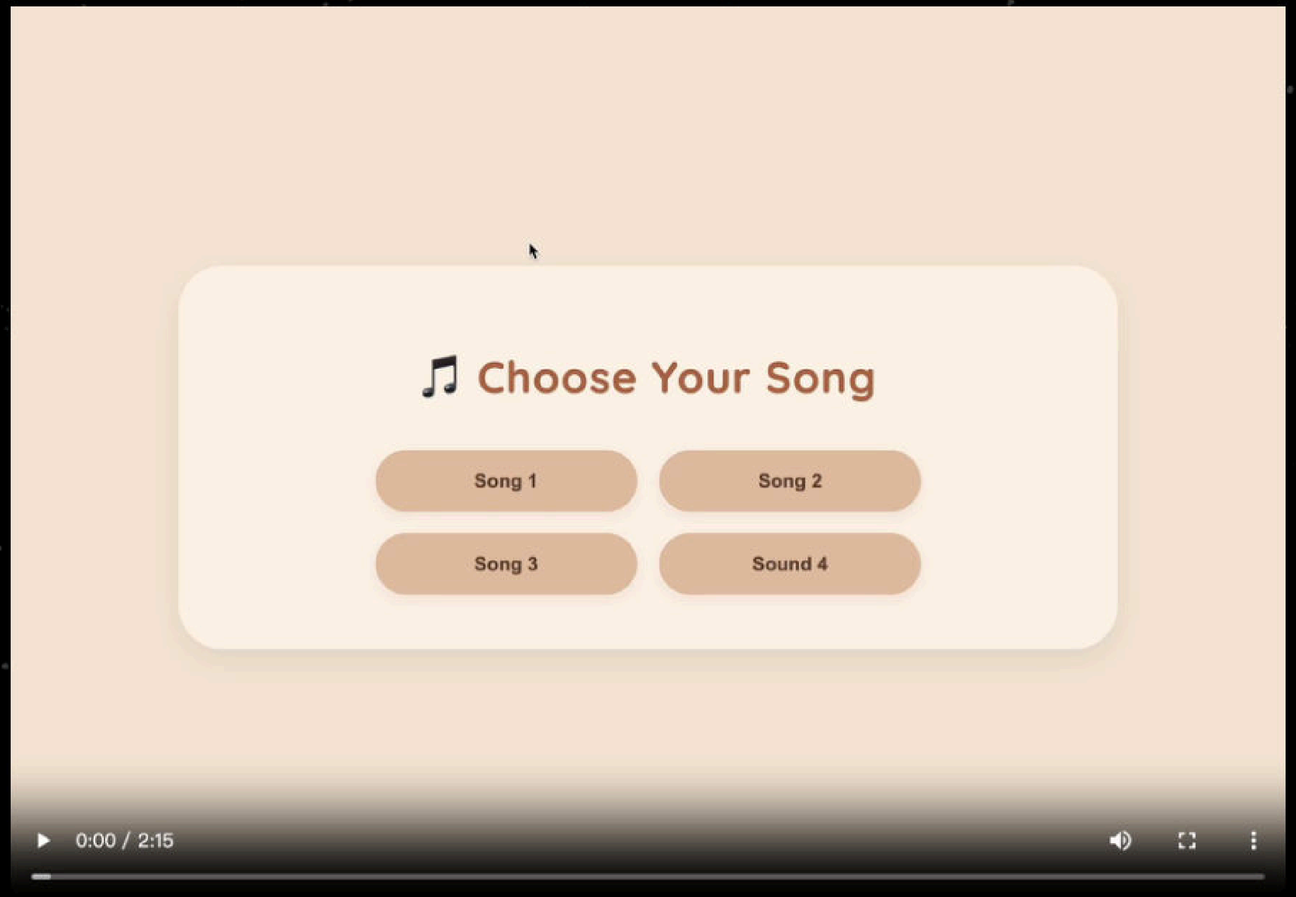
Activity 2: Learning p5.js Sound Visualization (Week 7)

Objective: Utilize p5.js to create foundational visual effects reflecting audio responses.
Process:Through p5.js learning, multiple visual effects were developed.
Result: While demonstrating feasibility, the visuals proved too monotonous to convey the richness of sound.

I have gone through an exploration process from simple to complex. At first, I could only control the size of the circle by sound, but I found the visual effect too monotonous. Then I attempted to create spectrograms and 3D geometric figures, gradually enriching the visual language. The greatest breakthrough was inspired by particle synthesis and began to experiment with "particle effects".

Formstorming: Final Project 2 Design

http://zhou37.phoenix.sheridanc.on.ca/DESN22848/Sound_Header_Template/



(Due to software issues, the video cannot be imported into Figma.)

This part presents a P5 Interactive Audio Web Header Portfolio. It expresses diverse visual forms through four distinct environments. From classic 2D spectrograms and radial pulse waves to dynamic 3D kinetic cube spaces, all visual effects are sound-driven. Geometric patterns can be manually generated and transformed in real-time through microphone input, breathing and morphing freely. Built with p5.js, this project delves into the synergistic relationship between sound frequencies, dynamic graphics, and user creativity, delivering an immersive interactive digital experience.

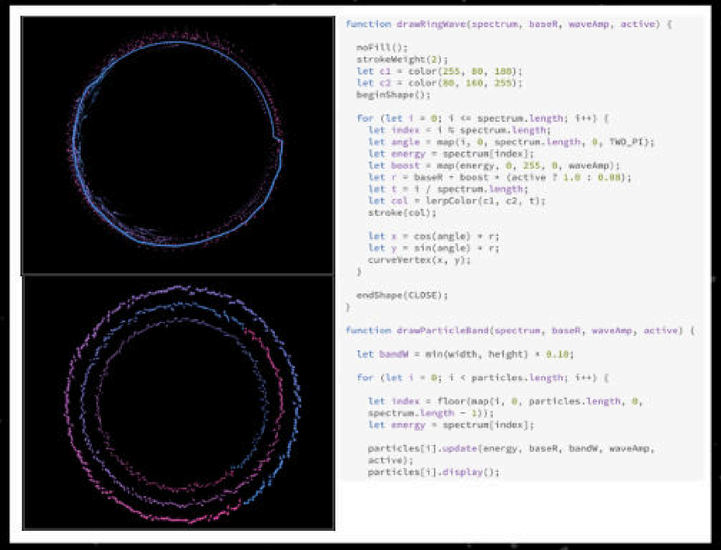
If the video below does not play, you can watch it on YouTube.
YOUTUBE: https://youtu.be/AmP7iFSQApk

Reflection: What I Learned

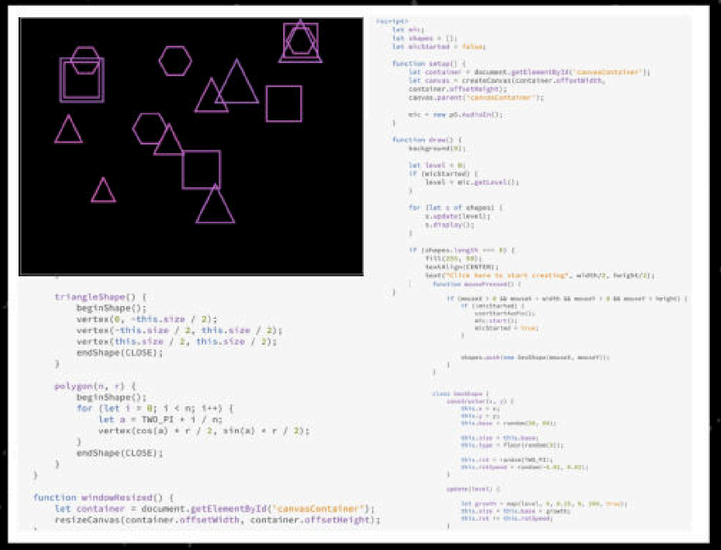


I learned a great deal from this project.

Starting with the p5.js code library, I gradually built the code step by step. From initially only being able to control the size of circles, to later creating spectrograms, and then experimenting with geometric shapes. I progressively grasped the meaning of “sound as data.” Concepts like amplitude, frequency, and waveform moved beyond theoretical abstractions and became tangible, manipulable visual effects.



Secondly, I believe my greatest takeaway was grasping the “layered nature of mapping.” While the initial simple circles demonstrated technical feasibility, they felt overly monotonous. Through multiple iterations, I discovered the most effective design approach wasn't cramming every sound parameter into the visuals, but rather setting 1-2 variations for each visual environment.



For instance, in “sound 4,” users actively shape the graphics by tapping the screen, and the visuals evolve through their interactions. These elements intuitively convey “what the sound is controlling.” This taught me that the core of interaction design isn't showcasing technical complexity, but ensuring everything remains sufficiently clear.

Reflection: Challenges



The biggest challenge was the “domino effect” of code debugging. This was the most frustrating yet rewarding experience throughout the entire project.

While developing the 3D cube visual effects, I encountered a classic case of “robbing Peter to pay Paul”: the particle system code ran smoothly, but when I added the FFT functionality for spectral analysis, the particle movement suddenly stuttered. After spending an entire day optimizing particle performance to achieve smooth motion, I discovered the spectrum's color mapping had completely failed. Fixing the color mapping then rendered the microphone input unusable... Each time I modified one component, another previously functional part would break.

This chain reaction made me keenly realize: in interactive systems, all modules are interconnected. Sound input, data analysis, visual rendering, and user interfaces—features that appear independent are actually tightly coupled within the code.

Reflection: Summary, Future Improvements

```
function setup() {
    let container = document.getElementById('canvasContainer');
    let canvas = createCanvas(container.offsetWidth,
    container.offsetHeight);
    canvas.parent('canvasContainer');

    fft = new p5.FFT(0.8, 128);
    fft.setInput(song);
}

function draw() {
    background(0);

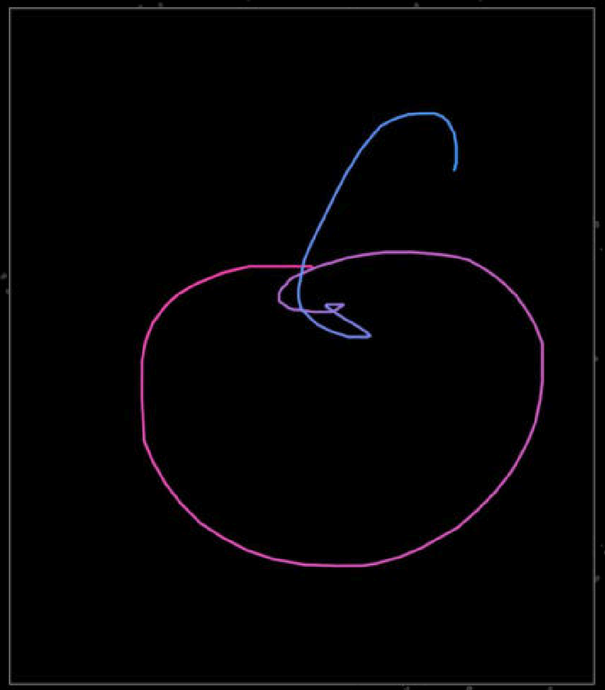
    if (isPlaying) {
        let spectrum = fft.analyze();
        noStroke();

        let barWidth = width / spectrum.length;
        let c1 = color(255, 80, 180);
        let c2 = color(80, 160, 255);

        for (let i = 0; i < spectrum.length; i++) {
            let value = spectrum[i];
            let h = map(value, 0, 255, 0, height);
            let t = i / (spectrum.length - 1);
            let c = lerpColor(c1, c2, t);

            fill(c);
            rect(i * barWidth, height - h, barWidth - 1, h);
        }
    }
}
```





Throughout this project, I gained comprehensive experience in sound visualization design—from theoretical concepts to practical implementation. This project has taught me that the essence of interaction design lies not in technology itself, but in how we leverage technology to establish meaningful, emotionally resonant dialogues between people and media.

Citations: Image

https://www.billviola.com/Viola%202%20bio.jpg
https://i.guim.co.uk/img/media/57b17423b7936bdac635c0e63fab112e99fbac6d/0_51_1917_1150/master/1917.jpg?width=1200&height=900&quality=85&auto=format&fit=crop&s=aa8877dfba6ea1a4f8ab4494059e0d7d
https://jonathangriffin.org/wp-content/uploads/2014/04/bill-viola.jpg
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https://img-blog.csdnimg.cn/img_convert/48de3197d29b02ae5a225ff79bd64411.png

Citations: Teaching Websites and Artist research Citation

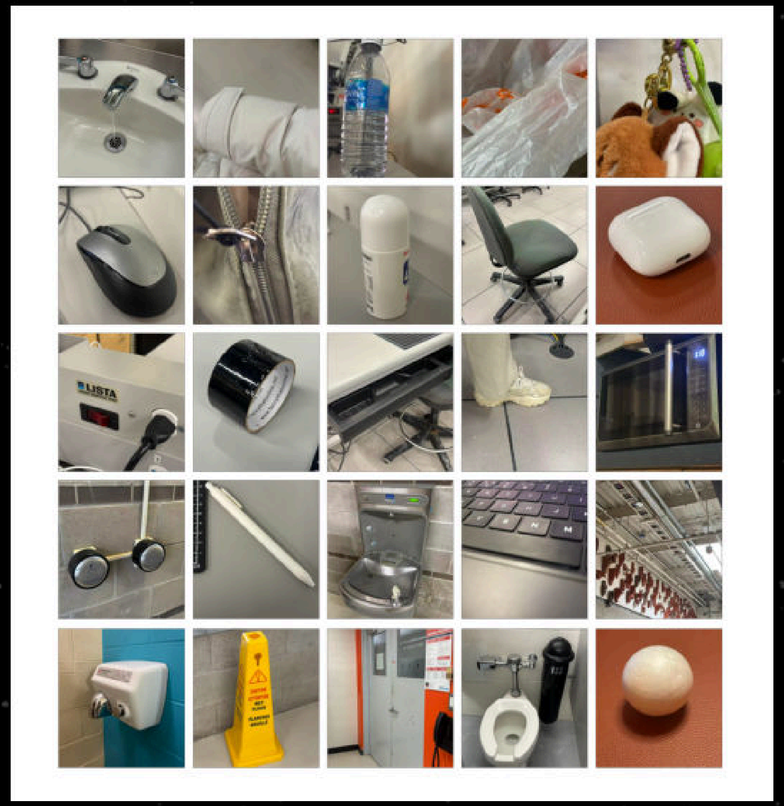
https://archive.p5js.org/zh-Hans/libraries/#content
https://p5js.org/zh-Hans/contribute/creating_libraries/#code
https://bkimg.cdn.bcebos.com/pic/b999a9014c086e061d95f18009546cf40ad163d95df1?x-bce-process=image/format,f_auto/quality,Q_70/resize,m_lfit,limit_1,w_536
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https://blog.csdn.net/gitblog_00240/article/details/152401040
https://webptogif.net/zh/mov-to-gif
https://granularsynthesis.info/MODELL-5

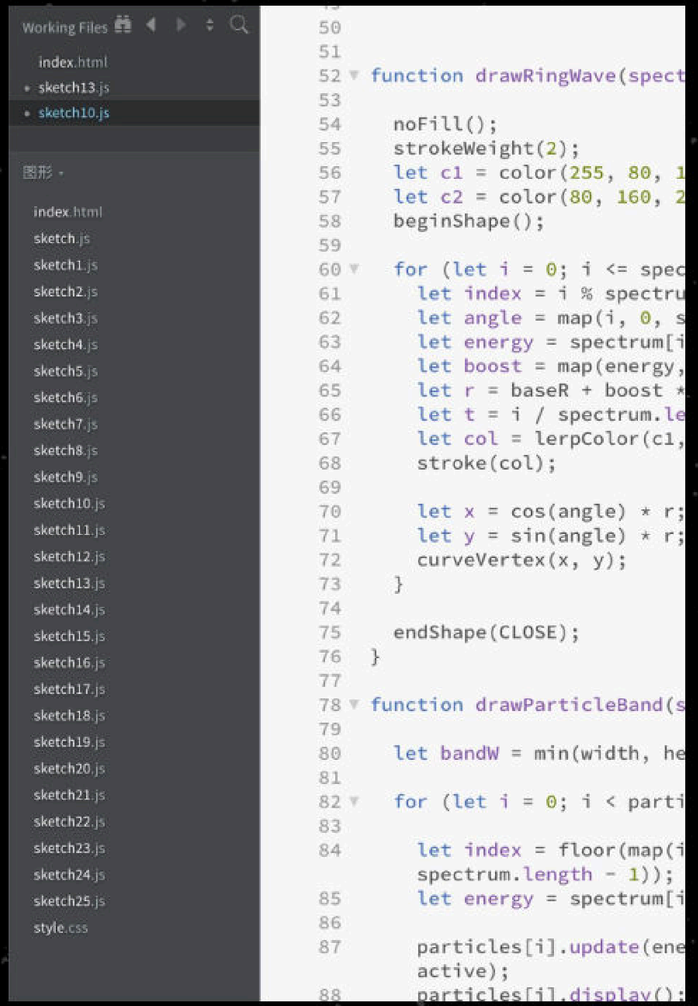
Porcel, V. (2009). Conversation with Bill Viola. IEMed.
Buffalo AKG Art Museum. (1997). Bill Viola: The Messenger.
Spielmann, Y. Model 5. See This Sound.
Ars Electronica. (1998). Granular And Synthesis.
IFFR. (2008). Modell 5.
Citation of course materials

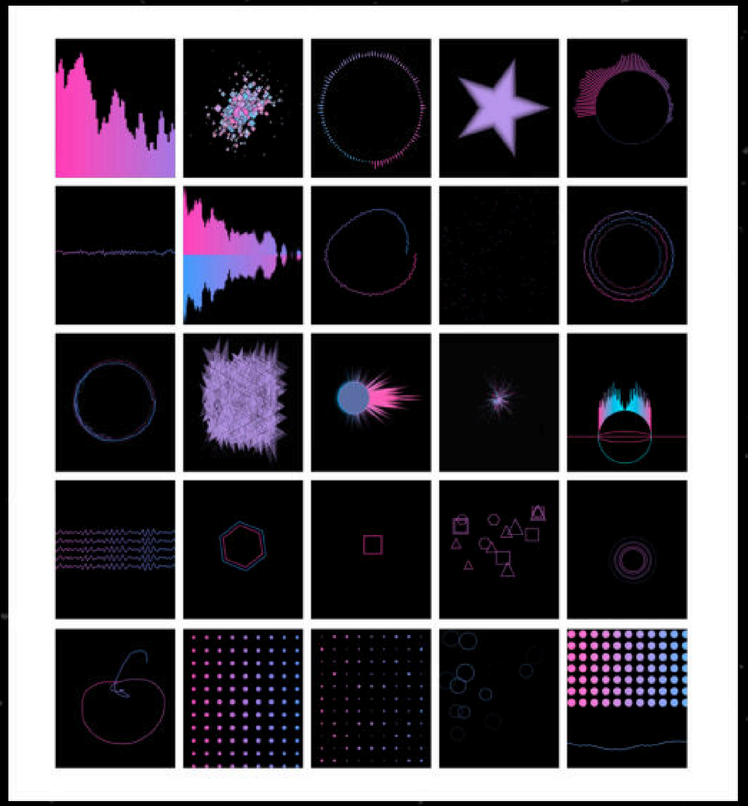
Citations: Cite the courseware

DESN22848 Week 6 Presentation: "Hear: Formstorming and Documentation", Sheridan College, Professor Steve Hudak
DESN22848 Week 6 courseware page 13: "Interpretation/Experience/Perception" three layers framework data
Pages 23-35 of DESN22848 Week 6 Courseware: Video Art History Cases
Pages 24-25 of DESN22848 Week 7 Courseware: Accessibility and P5.js Sound Library Application
DESN22848 Week 8 Courseware pages 50-56: Final Delivery Requirements for Project 2

Appendix: WEEK6-7

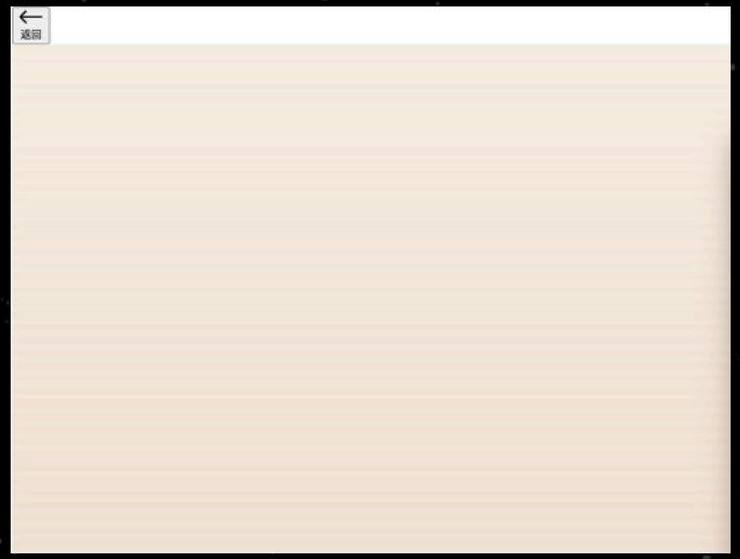






Appendix: WEEK8-9

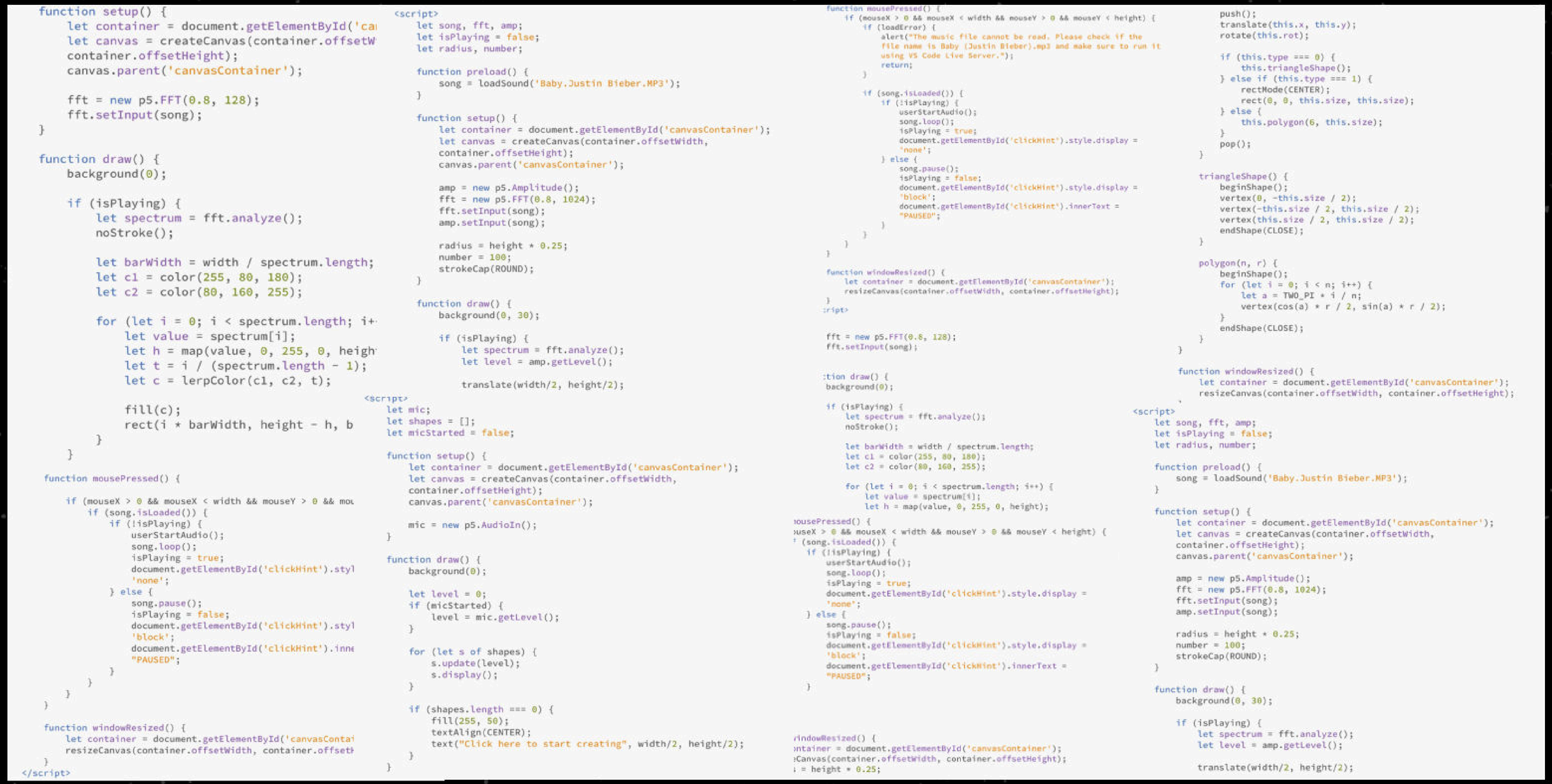
The bugs that occurred and the modified versions:







Appendix: WEEK8-9



```
function setup() {
    let container = document.getElementById('ca
    let canvas = createCanvas(container.offsetW
    container.offsetHeight);
    canvas.parent('canvasContainer');

    fft = new p5.FFT(0.8, 128);
    fft.setInput(song);
}

function draw() {
    background(0);

    if (isPlaying) {
        let spectrum = fft.analyze();
        noStroke();

        let barWidth = width / spectrum.length;
        let c1 = color(255, 80, 180);
        let c2 = color(80, 160, 255);

        for (let i = 0; i < spectrum.length; i+
            let value = spectrum[i];
            let h = map(value, 0, 255, 0, heigh
            let t = i / (spectrum.length - 1);
            let c = lerpColor(c1, c2, t);

            fill(c);
            rect(i * barWidth, height - h, b
        }
    }

function mousePressed() {

    if (mouseX > 0 && mouseX < width && mouseY > 0 && mou
        if (song.isLoaded()) {
            if (!isPlaying) {
                userStartAudio();
                song.loop();
                isPlaying = true;
                document.getElementById('clickHint').styl
                'none';
            } else {
                song.pause();
                isPlaying = false;
                document.getElementById('clickHint').styl
                'block';
                document.getElementById('clickHint').inne
                "PAUSED";
            }
        }
    }
}

function windowResized() {
    let container = document.getElementById('canvasConta
    resizeCanvas(container.offsetWidth, container.offset
}
</script>
```

```
<script>
    let song, fft, amp;
    let isPlaying = false;
    let radius, number;

    function preload() {
        song = loadSound('Baby.Justin Bieber.MP3');
    }

    function setup() {
        let container = document.getElementById('canvasContainer');
        let canvas = createCanvas(container.offsetWidth,
        container.offsetHeight);
        canvas.parent('canvasContainer');

        amp = new p5.Amplitude();
        fft = new p5.FFT(0.8, 1024);
        fft.setInput(song);
        amp.setInput(song);

        radius = height * 0.25;
        number = 100;
        strokeCap(ROUND);
    }

    function draw() {
        background(0, 30);

        if (isPlaying) {
            let spectrum = fft.analyze();
            let level = amp.getLevel();

            translate(width/2, height/2);
```

```
<script>
    let mic;
    let shapes = [];
    let micStarted = false;

    function setup() {
        let container = document.getElementById('canvasContainer');
        let canvas = createCanvas(container.offsetWidth,
        container.offsetHeight);
        canvas.parent('canvasContainer');

        mic = new p5.AudioIn();
    }

    function draw() {
        background(0);

        let level = 0;
        if (micStarted) {
            level = mic.getLevel();
        }

        for (let s of shapes) {
            s.update(level);
            s.display();
        }

        if (shapes.length === 0) {
            fill(255, 50);
            textAlign(CENTER);
            text("Click here to start creating", width/2, height/2);
        }
    }
```

```
function mousePressed() {
    if (mouseX > 0 && mouseX < width && mouseY > 0 && mouseY < height) {
        if (loadError) {
            alert("The music file cannot be read. Please check if the
            file name is Baby (Justin Bieber).mp3 and make sure to run it
            using VS Code Live Server.");
            return;
        }

        if (song.isLoaded()) {
            if (!isPlaying) {
                userStartAudio();
                song.loop();
                isPlaying = true;
                document.getElementById('clickHint').style.display =
                'none';
            } else {
                song.pause();
                isPlaying = false;
                document.getElementById('clickHint').style.display =
                'block';
                document.getElementById('clickHint').innerText =
                "PAUSED";
            }
        }
    }
}

function windowResized() {
    let container = document.getElementById('canvasContainer');
    resizeCanvas(container.offsetWidth, container.offsetHeight);
}
</script>

fft = new p5.FFT(0.8, 128);
fft.setInput(song);


tion draw() {
background(0);

if (isPlaying) {
    let spectrum = fft.analyze();
    noStroke();

    let barWidth = width / spectrum.length;
    let c1 = color(255, 80, 180);
    let c2 = color(80, 160, 255);

    for (let i = 0; i < spectrum.length; i++) {
        let value = spectrum[i];
        let h = map(value, 0, 255, 0, height);

ousePressed() {
ouseX > 0 && mouseX < width && mouseY > 0 && mouseY < height) {
 (song.isLoaded()) {
  if (!isPlaying) {
      userStartAudio();
      song.loop();
      isPlaying = true;
      document.getElementById('clickHint').style.display =
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      song.pause();
      isPlaying = false;
      document.getElementById('clickHint').style.display =
      'block';
      document.getElementById('clickHint').innerText =
      "PAUSED";
  }


indowResized() {
ontainer = document.getElementById('canvasContainer');
Canvas(container.offsetWidth, container.offsetHeight);
 = height * 0.25;
```

```
        push();
        translate(this.x, this.y);
        rotate(this.rot);

        if (this.type === 0) {
            this.triangleShape();
        } else if (this.type === 1) {
            rectMode(CENTER);
            rect(0, 0, this.size, this.size);
        } else {
            this.polygon(6, this.size);
        }
        pop();
    }

    triangleShape() {
        beginShape();
        vertex(0, -this.size / 2);
        vertex(-this.size / 2, this.size / 2);
        vertex(this.size / 2, this.size / 2);
        endShape(CLOSE);
    }

    polygon(n, r) {
        beginShape();
        for (let i = 0; i < n; i++) {
            let a = TWO_PI * i / n;
            vertex(cos(a) * r / 2, sin(a) * r / 2);
        }
        endShape(CLOSE);
    }
}

function windowResized() {
    let container = document.getElementById('canvasContainer');
    resizeCanvas(container.offsetWidth, container.offsetHeight);
}

<script>
    let song, fft, amp;
    let isPlaying = false;
    let radius, number;

    function preload() {
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        canvas.parent('canvasContainer');

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        fft = new p5.FFT(0.8, 1024);
        fft.setInput(song);
        amp.setInput(song);

        radius = height * 0.25;
        number = 100;
        strokeCap(ROUND);
    }

    function draw() {
        background(0, 30);

        if (isPlaying) {
            let spectrum = fft.analyze();
            let level = amp.getLevel();

            translate(width/2, height/2);
```

THANK YOU